Ziseok Lee

Personal Website: https://ziseoklee.github.io/

Google Scholar: https://scholar.google.com/citations?user=C3x_tewAAAAJ&hl=en

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Integrated Masters with PhD Program

Department of Biomedical Sciences

Seoul National University

Research Interests

Generative Models, AI for Science, Computer Vision

Education

Seoul National University — 2025.03 - present

Masters-PhD Program in Biomedical Sciences advised by Prof. Kyungsu Kim

Seoul National University — 2021.03 - 2025.02

B.S. in Mathematical Sciences (Major), B.S. in Computer Science and Engineering (Double Major)

GPA 4.22/4.30 (Summa Cum Laude)

Research Experience

▼ Student Researcher at **AIBL** — 2024.10 - present

Probabilistic Generative Models for Biomolecule Generation and Drug Design

Designed stochastic interpolant-based sampling methods for drug optimization and integrating physics-aware constraints for structure-based drug design.

Derived and implemented the ACE framework for multi-expert diffusion steering, fixing Marginal Path Collapse via the Path Existence Criterion.

Authored a grant proposal and secured funding for a 3-year international joint research project between SNU and Harvard Medical School (Prof. Kwang-Soo Kim).

Co-first authored multiple papers on applications of diffusion models in computer vision and molecular modeling.

▼ Undergraduate Research Assistant at **SNU MLLAB** — 2024.07 - 2024.09

Blackbox Optimization

▼ Undergraduate Research Assistant at **CTA Lab** — 2024.01 - 2024.08

NP-hard Subgraph Matching Algorithms with Safety Conditions for Reduced Candidate Space

▼ Undergraduate Research Assistant at **HYKE Group** — 2022.12 - 2023.07

Collective Motion in the Vicsek Particle Model, Hyperbolic Conservation Laws with Nonlocal Relaxation

Awards & Scholarships

AI Seoul Tech Graduate Scholarship (Seoul Scholarship Foundation)	2025
B.S. Degree Honors: Summa Cum Laude (Seoul National University)	2025
Presidential Science Scholarship (Korea Student Aid Foundation)	2023-2024
Dean's List (Department of Mathematical Sciences, SNU)	2023-2024

Professor Heo Sik Scholarship (Department of Mathematical Sciences, SNU) 2022

Gwanak Foundation Scholarship (Gwanak Foundation) 2021

Teaching Experience

▼ Teaching Assistant for **[Data Mining and Machine Learning]** 2026.03 - 2026.06

Assisted Professor Kyungsu Kim in teaching a third-year undergraduate course, supporting students through discussion sessions.

▼ Teaching Assistant for **[Introduction to Data Science]** 2025.03 - 2025.06

Assisted Professor Kyungsu Kim in teaching a second-year undergraduate course, supporting students through discussion sessions.

▼ Teaching Assistant for **[Calculus 1]** 2024.03 - 2024.06

Taught a course on Calculus 1. Was awarded **Excellent TA** and gave a case presentation.

▼ Student-Directed Seminar **[Understanding the Brain as a Complex System]** 2023.09 - 2023.12

Organized and taught a student-directed seminar integrating theoretical neuroscience, network science, deep learning, biology, and psychology to understand the brain as a complex, dynamical, entangled system.

▼ Peer Tutoring **[Calculus Tutoring for First Year Students]** 2023.03 - 2023.06

Taught a tutoring class for "Differential and Integral Calculus 1" during the Spring semester of 2023.

Grants & Funding

▼ Strategic Hub for International Research Collaboration (SNU Office of Research Affairs) 2026 - 2028

Project Title: An International Research Hub for Developing a 4D Molecular Generative AI Platform to Control Dynamic Protein Interaction Networks.

Funding: ~$380k (3 Years)

Description: Authored the grant proposal and secured funding for a 3-year international joint research project between SNU and Harvard Medical School (Prof. Kwang-Soo Kim).

My Role: I lead the design of stochastic interpolant-based diffusion models for drug optimization and integrating physics-aware constraints for structure-based drug design.

Publications

* Equal contribution, † Corresponding Author

Conference Papers

[Co-First & Co-Corresponding Author] *On the Collapse of Generative Paths: A Criterion and Correction for Diffusion Steering*, Ziseok Lee*†, Minyeong Hwang*, Wooyeol Lee, Sanghyun Jo, Jihyung Ko, Young Bin Park, Jae-Mun Choi, Eunho Yang†, Kyungsu Kim†, Forty-third International Conference on Machine Learning (ICML) (2026)

[Co-first Author] *TRACE: Your Diffusion Model is Secretly an Instance Edge Detector*, Sanghyun Jo*, Ziseok Lee*, Wooyeol Lee, Jonghyun Choi, Jaesik Park†, Kyungsu Kim†, The Fourteenth International Conference on Learning Representations **(ICLR Oral, top 1.1%)** (2026)

[Co-first Author] *Early Timestep Zero-Shot Candidate Selection for Instruction-Guided Image Editing*, Joowon Kim*, Ziseok Lee*, Donghyeon Cho, Sanghyun Jo, Yeonsung Jung, Kyungsu Kim†, Eunho Yang†, Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) (2025)

Workshop Papers

[Coauthor] *HybridLinker: Topology-Guided Posterior Sampling for Enhanced Diversity and Validity in 3D Molecular Linker Generation*, Minyeong Hwang, Ziseok Lee, Kwang-Soo Kim, Kyungsu Kim†, Eunho Yang†, ICML 2025 Generative AI and

Biology (GenBio) Workshop (2025)

Preprints

[Co-first Author] *Value-Informed Schedule Trimming for Accelerated Sequential Monte Carlo Guidance in Discrete Diffusion*, Jaehyeon Kim*, Ziseok Lee*, Siyoung Choi, Kyungsu Kim†, (2026)

[Co-first Author] *ISAC: Training-Free Instance-to-Semantic Attention Control for Improving Multi-Instance Generation*, Sanghyun Jo*, Wooyeol Lee*, Ziseok Lee*, Kyungsu Kim†, arXiv preprint arXiv:2505.20935 (2025)

Academic Services

[2026.05] Reviewer for NeurIPS 2026.

[2025.04] Assisted Professor Kyungsu Kim in reviewing submissions in the field of computer vision for ICCV 2025.

[2025.03] Assisted Professor Kyungsu Kim in reviewing two submissions in the field of flow matching for ICML 2025.

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, Scikit-learn, OpenCV

Tools & Platforms: High-performance GPU server management, Linux system administration

Languages: Korean (native), English (fluent)

References

Dr. Kyungsu Kim (Primary advisor)

Professor, School of Transdisciplinary Innovations

Seoul National University

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